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Revision — the circle and its measures

Circumference, area, arcs, sectors and compound shapes on one page.

LESSON 107–108 | 2 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 8 AND 15 CONSOLIDATION

LESSON CLOCK

20' | 25' | 25' | 8' | 2

The two formulae	20 min
Fractions of a circle	25 min
Compound shapes	25 min
The errors that remain	8 min
Homework	2 min

LEARNING OBJECTIVES

- Choose between $C = 2\pi r$ and $S = \pi r^2$ from the wording.
- Work forwards and backwards with both formulae.
- Find arc lengths and sector areas as fractions.
- Handle a compound shape confidently.

TEXTBOOK

National	pp. 314–318
Cambridge	Stage 7, pp. 80–84, 148–156
Workbook	Revision 15

01 Explanation

The two formulae

WANTED	FORMULA	UNITS	BACKWARDS
circumference	$C = 2\pi r = \pi d$	cm	$r = C \div 2\pi$
area	$S = \pi r^2$	cm^2	$r = \sqrt{(S \div \pi)}$

R	C	S
7	44	154
14	88	616
21	132	1386

THE UNITS DECIDE THE FORMULA

Fencing, edging and string are lengths; turf, glass and paint are areas. Reading the material tells you which formula before any number is written.

Fractions of a circle

FIGURE	ARC OR CURVE	AREA	PERIMETER
semicircle, $r = 7$	22	77	36
quadrant, $r = 7$	11	38.5	25
$120°$ sector, $r = 21$	44	462	86

AREA IS A CLEAN FRACTION; PERIMETER IS NOT

A semicircle has half the area of the disc but not half the perimeter — the straight edge is added. This is the single most repeated error of the chapter.

Compound shapes

SHAPE	AREA	PERIMETER
a semicircle of radius 7 on a 14×10 rectangle	217	56
a square of side 14 less a disc of radius 7	42	100
a ring between radii 7 and 14	462	132

SKETCH, LABEL, THEN CALCULATE

Writing the area of each piece on the sketch, and tracing the boundary with a finger, catches almost every error before it happens.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
diameter used as radius	$\pi \cdot 14^2$	$\pi \cdot 7^2$
$(\pi r)^2$	484	$\pi r^2 = 154$
semicircle perimeter without the diameter	22	36
square root forgotten	$r = 49$	$r = 7$
units missing or wrong	154 *cm*	154 *cm*^{2}

02 Worked examples

EXAMPLE 1 **A circle has radius *21 cm*. Find its circumference and area.**

1. $C = 2 \cdot \frac{22}{7} \cdot 21 = 132$ cm.
2. $S = \frac{22}{7} \cdot 441 = 1386$ cm^2.
3. One in *cm*, one in *cm*^{2}.

ANSWER

132 *cm* and 1386 *cm*^{2}

EXAMPLE 2 **A *120°* sector has radius *21 cm*. Find its arc, area and perimeter.**

1. Fraction: $\frac{1}{3}$.
2. Arc: $132 \div 3 = 44$ cm.
3. Area: $1386 \div 3 = 462$ cm².
4. Perimeter: $44 + 42 = 86$ cm.
   Two radii added.

ANSWER

44 cm, *462 cm²*, *86 cm*

EXAMPLE 3 **A disc has area *1386 cm²*. Find its circumference.**

1. $r^2 = 1386 \div \frac{22}{7} = 441$
2. $r = 21$ cm.
   The square root — easily forgotten.
3. $C = 2 \cdot \frac{22}{7} \cdot 21 = 132$ cm.

ANSWER

132 cm

03 Interactive model

Give six questions and ask only whether each needs C or S; sorting them takes two minutes and removes most of the errors in the control work.

Length or area?

Fencing round a circular pond needs:

- C
- S
- both
- neither

Turf for the same pond area needs:

- C
- S
- both
- neither

With $r = 21$, C is:

- 66
- 132
- 1386
- 441

With $r = 21$, S is:

- 132
- 441
- 1386
- 66

A semicircle of radius 7 has perimeter:

- 22
- 36
- 44
- 77

Its area is:

- 22
- 36
- 77
- 154

An area of 1386 gives r equal to:

- 21
- 441
- 132
- 66

154 cm as an answer for an area is:

- fine
- the wrong units
- a radius
- a perimeter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circumference	Aylana uzunligi	Длина окружности
Area	Yuza	Площадь
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Compound shape	Murakkab shakl	Составная фигура
Units	O‘lchov birligi	Единицы измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

C is measured in:

- cm
- cm^2
- cm^3
- degrees

S is measured in:

- cm
- cm^2
- cm^3
- degrees

A disc of radius 14 has area:

- 88
- 308
- 616
- 196

A semicircle of radius 14 has perimeter:

- 44
- 72
- 88
- 28

A $120°$ sector is what fraction?

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{4}$
- $\frac{1}{6}$

To find r from the area you:

- divide by π
- divide by π then take the root
- divide by 2π
- take the root

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 7$: C

   ANSWER 44 cm

2. $r = 7$: S

   ANSWER 154 cm^2

3. $r = 14$: C

   ANSWER 88 cm

4. $r = 14$: S

   ANSWER 616 cm^2

5. $r = 21$: C

   ANSWER 132 cm

6. $r = 21$: S

   ANSWER 1386 cm^2

7. Units for an area

   ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. A semicircle of radius 7: area and perimeter

   ANSWER $77\ cm^2, 36\ cm$

2. A quadrant of radius 7

   ANSWER $38.5\ cm^2, 25\ cm$

3. A $120°$ sector of radius 21

   ANSWER $462\ cm^2, 86\ cm$

4. $C = 88$: the radius

   ANSWER 14

5. $S = 616$: the radius

   ANSWER 14

6. A ring between radii 7 and 14: the area

   ANSWER $462\ cm^2$

7. A square of side 14 less a disc of radius 7

   ANSWER $42\ cm^2$

Hard · stretch and reason

7 PROBLEMS

1. A disc of area 1386: its circumference

   ANSWER $132\ cm$

2. A disc of circumference 132: its area

   ANSWER $1386\ cm^2$

3. A semicircle of radius 7 on a 14×10 rectangle: area and perimeter

   ANSWER $217\ cm^2$, $56\ cm$

4. A $60°$ sector of radius 14: the area

   ANSWER $102.67\ cm^2$

5. A ring between radii 7 and 14: its outer and inner boundary total

   ANSWER $132\ cm$

6. Doubling the radius: what happens to C and S?

   ANSWER C doubles, S quadruples

7. Why is $154\ cm$ wrong for an area?

   ANSWER Areas need square units

07 Homework

Homework — 5 tasks

Write the units first, and let them tell you which formula to use.

1. Find C and S for circles of radius $3.5\ cm$ and $28\ cm$.
2. A circle has circumference $66\ cm$. Find its radius and area.
3. Find the area and perimeter of a semicircle of radius $14\ cm$.
4. Find the area of a $45°$ sector of a circle of radius $28\ cm$.

5. A rectangle 20×14 cm has a semicircle of radius 7 cm cut from one short side. Find the area left.